

Final Project: Audio Equalizer

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1. Abstract

This project focuses on the design and construction of an audio equalizer. The main objective was to create a circuit capable of adjusting sound frequencies across three ranges: low (up to 320 Hz), high (above 3200 Hz), and mid-range (from 320Hz to 3200 Hz). This involved the use of a high pass, low pass, and mid-pass filters, combined with operational amplifiers (op-amps) for signal inversion, and a power amplifier. Methods included rigorous testing with oscilloscopes, revisiting lab-derived equations, and conducting extensive research to achieve the desired signal modifications when combining the filters—a step beyond our lab's scope, where filters were only studied in isolation.

The key results were promising: sound output from the speaker was achieved, and frequency response tests displayed the filters' effects on various frequencies. Those images showed the mid-pass, high-pass, and low-pass filter's frequency response, with behaviors consistent with theoretical expectations.

The relevance of this project extends to the broader context of modern devices, which commonly employ a range of amplifiers and equalizers for volume control and sound quality enhancement in speakers. Moreover, the in-depth exploration of op-amps provided critical insights—knowledge that is increasingly vital given their ubiquitous role in today's electronic devices. This project not only reinforced the practical application of electronic engineering principles but also underscored the importance of sound control technology in the current era.

2. Objectives

In this project we embarked on the ambitious task of conceptualizing, designing, and implementing a circuit that could adequately manage audio signals across three distinct frequency bands. Central to our mission was the development of a trio of filters – low pass, mid pass, and high pass – which were the foundation to modify the sound profile across the bass, midrange, and treble frequencies.

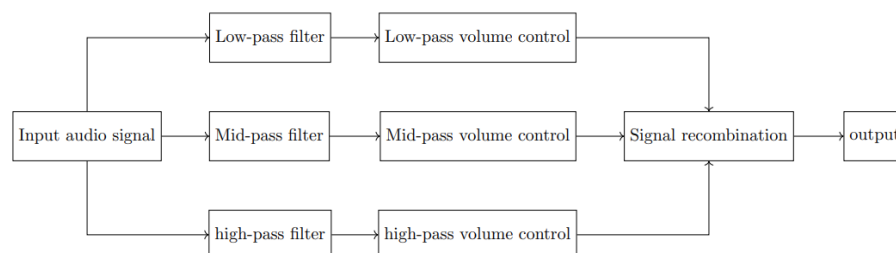


Figure 1: Audio equalizer framework

As we can observe in Figure 1, we laid out a clear process for our design, indicating the pathway of the audio signal through each filter and its corresponding volume control before merging in a unified output. We had certain constraints that we should follow. The low pass filter was tasked with attenuating frequencies above 320 Hz, essential for isolating the bass elements of the audio. The mid pass filter, serving as a band pass, was finely tuned to operate between 320 Hz and 3.2 kHz, a range that captures midrange tones. The treble frequencies, those above 3.2 kHz, were managed by the high pass filter, ensuring clarity and definition in the higher octaves of the sound spectrum.

Following the filters, we proceed with the amplifiers. We had to adjust things so that when all the equalizers were turned down, the voltage stayed under 15 mVRMS. When we turned everything up, the voltage needed to be around 100 mVRMS. We checked this at different pitches to make sure the sound stayed good at both low and high volumes.

As with the previous evaluation, we also analyzed the power of the amplifier, which was expected to exceed 400 mW across a frequency range from 200 Hz to 10 kHz. This parameter was essential for ensuring that our circuit could drive speakers effectively without distortion.

3. Theory

Before constructing the equalizer, we need to get a deeper understanding of all the concepts around audio, signals, filters, and amplifiers.

What is an equalizer?

In the world of sound systems, an equalizer is a vital tool that allows users to modify the sound by adjusting the balance of frequency components within an audio signal. It achieves this by amplifying (boosting) or attenuating (cutting) specific frequencies to modify the sound for the listener's preferences or the acoustic characteristics of an environment. The term frequency in this context refers to the number of times a wave repeats itself within a second and is measured in hertz (Hz). In sound, frequency is perceived as pitch, the higher the frequency, the higher the pitch of the sound.

When we talk about frequencies, we also need to understand the way it is perceived. Gain is the term we use to measure those frequencies and it is in 'decibels' (dB). Gain is a key concept to understand an equalizer; as Equation (1) shows, it represents the ratio of output to input voltage or amplitude (in some way the ratio of amplification on our circuit). The term 'decibels' quantifies this gain, providing a logarithmic way to express large ratios of power or voltage, which corresponds closely to how humans perceive sound.

$$\text{Equation (1): } dB = 20 \log_{10}\left(\frac{V_{out}}{V_{in}}\right)$$

Filters schematics

Going into the circuits of an equalizer, we employed three filters: Low-pass, high-pass, and mid-pass filter. For the sake of simplicity, we made use of passive filters, that do not require any external power to work. For low frequencies, the low pass filter is a very simple and common solution. As we can see in

Figure 2, it uses a resistor connected in series with a capacitor to ground, effectively allowing lower frequencies to pass while blocking higher frequencies. This setup takes advantage of the capacitor's property where its reactance decreases with increasing frequency, making it a natural barrier for high frequencies, isolating bass tones in the audio signal.

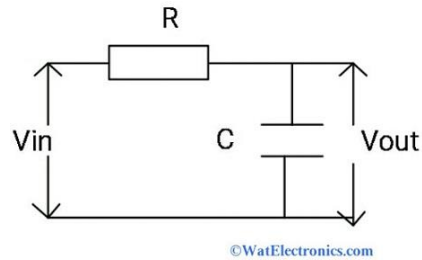


Figure 2: Passive low-pass filter

Following we have the high pass filter that employs a flipped configuration of the low pass filter (Figure 3), with the capacitor and resistor positions reversed. In this setup, the capacitor's role shifts to blocking low frequencies, while high frequencies pass through. This arrangement ensures that higher pitch sounds, or treble, are emphasized, filtering out lower frequency noise.

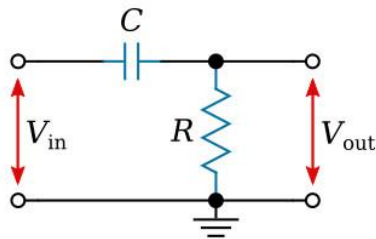


Figure 3: Passive high-pass filter

The mid pass filter, doubling as a band pass filter, combines the principles of both low and high pass filters using both an inductor and a capacitor (Figure 4). This dual-component setup creates a pathway for mid-range frequencies by blocking frequencies that are too high (thanks to the inductor) and too low (thanks to the capacitor). This selective filtering is crucial for maintaining the integrity of mid-range sounds like vocals and many instruments, which often carry the melody in music.

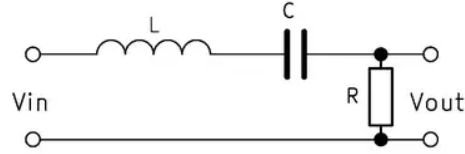


Figure 4: RLC band-pass filter

These filters, through their unique configurations of capacitors, resistors, and inductors, are crucial in defining the frequency response and overall functionality of the equalizer. By amplifying these filtered signals, we can finely tune the audio output, enhancing the listening experience.

Theory and equations for filters

After learning the basics of each filter, we need to understand how to implement them and make them filter the correct frequencies. For the low pass filter, the key equation involves the calculation of the cutoff frequency, which is determined by the values of the resistor and capacitor in the circuit. Equation (2) shows the final derivation of the relationship between gain and the circuit components and its variables are in reference of Figure 2.

$$\text{Equation (2): } f_c = \frac{1}{2\pi RC}$$

The high pass filter, though having a different arrangement of components, shares a similar foundational equation to determine its cutoff frequency (High Pass Filter Equation). We made use of Equation (2) however this time the variables are in relation to Figure 3.

For the band pass filter, the equation becomes more complex as it integrates aspects of both the low pass and high pass filters. Is crucial first to determine the range of frequencies that the filter allows to pass through since we will base our calculations on them. Equation (3) and Equation (4) show the relationship between the resistor, capacitor, and inductor.

$$\text{Equation (3): } \omega_{(c1)} = 2\pi f_{c1} = -\frac{R}{2L} + \sqrt{\left(\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}\right)}$$

$$\text{Equation (4): } \omega_{(c2)} = 2\pi f_{c2} = \frac{R}{2L} + \sqrt{\left(\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}\right)}$$

From these equations we can derive Equation (5) where we can finally get all the necessary components to build the desired band-pass filter.

$$\text{Equation (5): } \omega_{(c2)} - \omega_{(c1)} = \frac{R}{L}$$

Amplifiers and Signal inversion

In our audio equalizer project, operational amplifiers (op-amps) are essential for signal amplification (Figure 5). An op-amp amplifies the input signal by taking a small voltage difference between its two input terminals and producing a significantly larger voltage output.

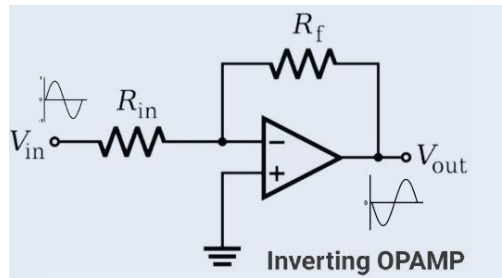


Figure 5: Inverting OP-AMP

This amplification is controlled by the feedback network, typically consisting of resistors, which determines the gain of the op-amp. By adjusting these resistors, we can set how much the signal is amplified. For the inverting op-amp configuration, Equation (6) can explain how we should set up the amplifier to get a desired voltage.

$$\text{Equation (6): } \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_{in}}$$

This capability of op-amps to boost signal strength is crucial, especially after the power loss from the filters. Additionally, the op-amps' ability to invert the signal ensures the output is in phase with the original input, a vital step for maintaining sound quality. The strategic placement of op-amps between the equalizers and the final amplifier in our design is key to conditioning the signal for optimal sound output before it is plugged to the speakers.

Integrated circuits

Now let's take the theory of the op-amp into real life circuits. We used the LM386 integrated circuits. They are a popular low-voltage audio power amplifier widely used in audio applications since they are the real-life representation of the op-amp. It's designed for battery-powered devices due to its low power consumption. It features a wide operating voltage range, typically from 4 volts to 12 volts. The

LM386 includes internal gain setting resistors, allowing users to set the gain with just a few external components, making it useful and easy to use in a variety of audio amplification applications. We can see its pin layout in Figure (6).

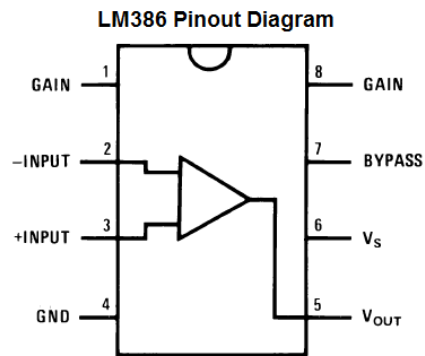


Figure 6: LM386 pin layout

4. Procedure

To construct our audio equalizer, we methodically worked through each component as shown in Figure (7).

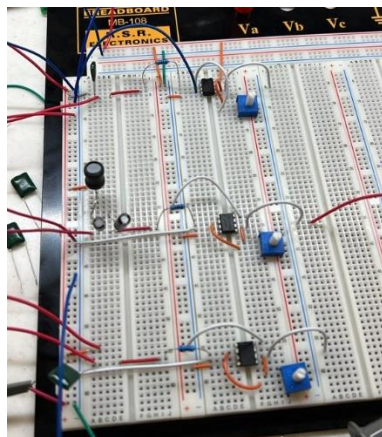


Figure 7: Filters with its respective filter amplifier

Beginning with the low pass filter, we applied the theoretical formula to calculate the needed resistor and capacitor values. We selected a 0.1 μF capacitor and rounded the resistor value to 4974 ohms, as determined by the formula. For the high pass filter, we utilized the same resistance value but opted for a 0.01 μF capacitor to meet the higher cutoff frequency of 3200 Hz, as opposed to the low pass cutoff of 320 Hz. In crafting the mid-pass filter, our calculations, based on the provided formulas, led us to choose a 24.2 μF capacitor, an 18-ohm resistor, and a 1mH inductor. This configuration was key to achieving the desired frequency range for this filter.

After establishing each filter, we focused on signal amplification. Using a 10k potentiometer in conjunction with various resistors, we fine-tuned the output from each filter. The circuit design ensured that all filters were arranged in parallel, with each one connected in series to a volume control mechanism, which we implemented using inverted op-amps and potentiometers for feedback resistance. This setup allowed for the individual adjustment of each frequency band's volume. Finally, as seen in Figure (8), the signals from each filter were combined at the signal recombination stage and passed through another inverted op-amp for overall volume control, before being fed into the power amplifier, built with an LM386 chip, to drive the output to the speakers. This detailed and sequential approach ensured the effective functionality of each component, culminating in a well-performing audio equalizer.

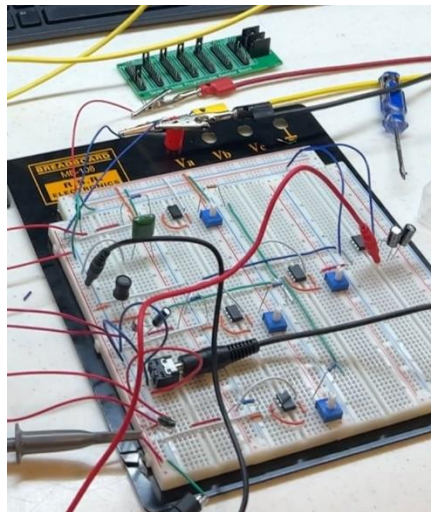


Figure 8: Final Equalizer and Amplifier

5. Results

In our results, we carefully assessed each filter's performance mentioned on the project constraints. The Frequency Response Analysis (FRA) for each filter - low pass, high pass, and mid pass - confirmed that they adhered to the specified cutoff frequencies. The low pass filter effectively limited frequencies

above 320 Hz (Figure 9), the high pass filter constrained those below 3200 Hz (Figure 10), and the mid pass filter accurately shaped the mid-range frequencies (Figure 11).

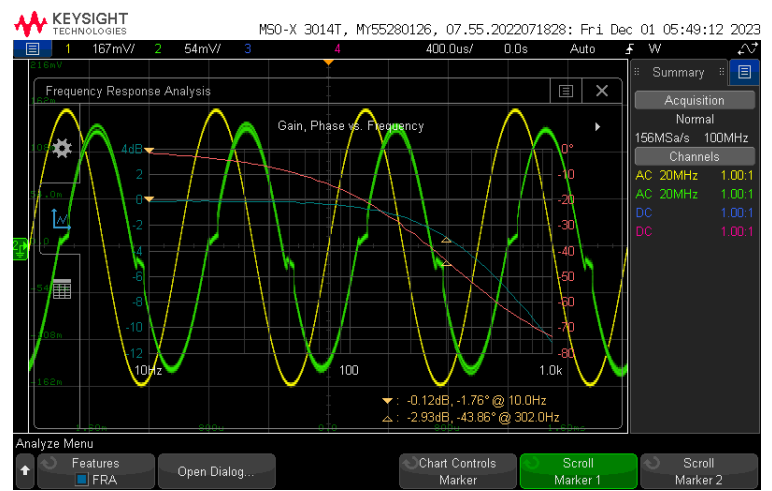


Figure 9: FRA plot for the low-pass filter

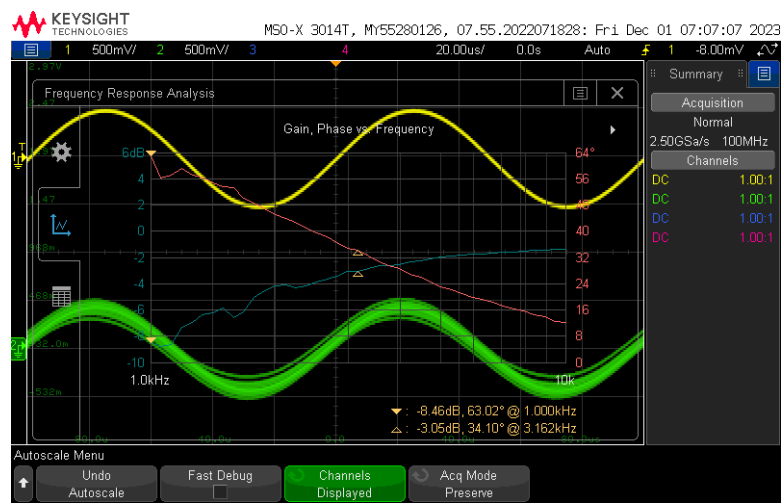


Figure 10: FRA plot for the high-pass filter

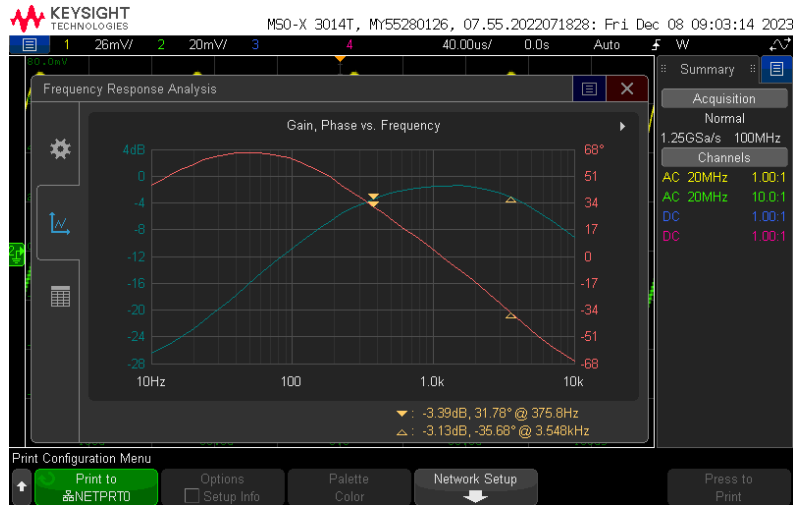


Figure 11: FRA plot for the band-pass filter

Continuing our testing, we measured the output voltage from the amplifier with the control knobs set to their minimum, as depicted in Figures 12, 13, and 14. These figures confirmed that the output voltage stayed within the desired range, effectively minimizing frequency amplification, and lowering the volume as intended.

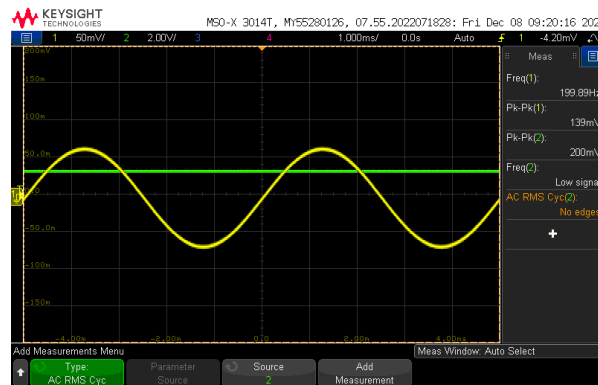


Figure 12: Oscilloscope image for minimum amplification at 200Hz

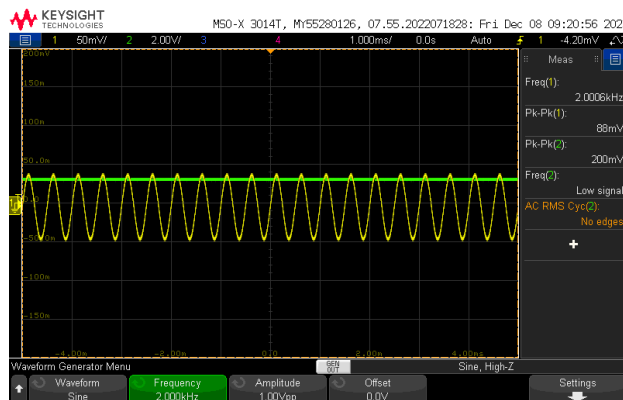


Figure 13: Oscilloscope image for minimum amplification at 2000Hz

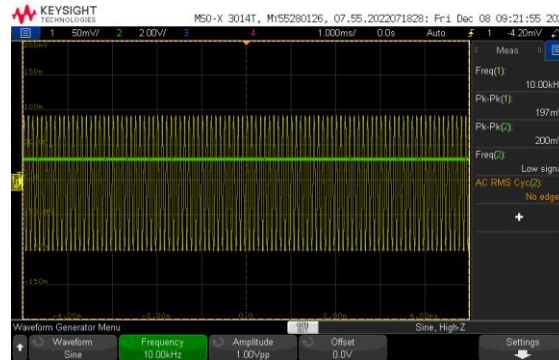


Figure 14: Oscilloscope image for minimum amplification at 10000Hz

We performed subsequent tests with the control knobs at maximum settings, however, they revealed some discrepancies. The output voltage at higher frequencies reached up to 2 Vpp, exceeding expected values of 0.1Vpp. Figures 15, 16, and 17 display oscilloscope readings at various frequencies, adjusted to maintain a reasonable voltage range, rather than at maximum knob settings.

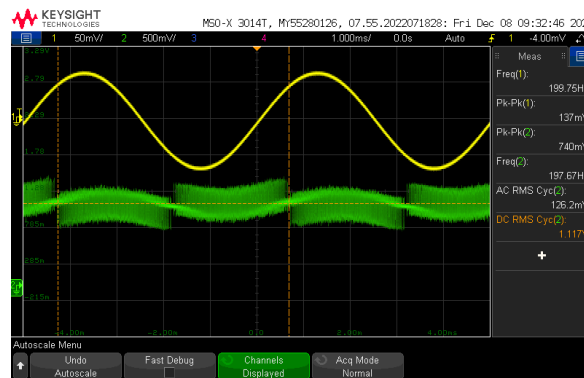


Figure 15: Oscilloscope image for maximum amplification at 200Hz

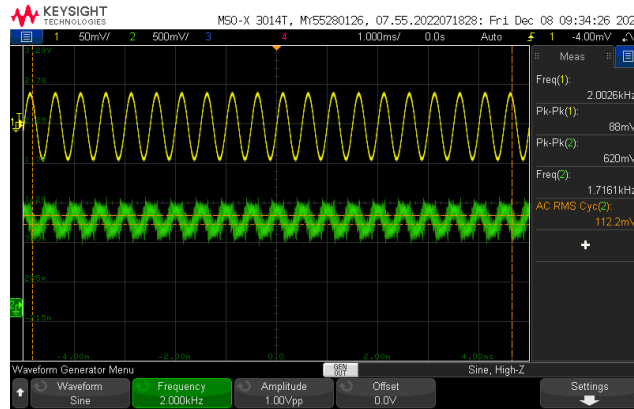


Figure 16: Oscilloscope image for minimum amplification at 2000Hz

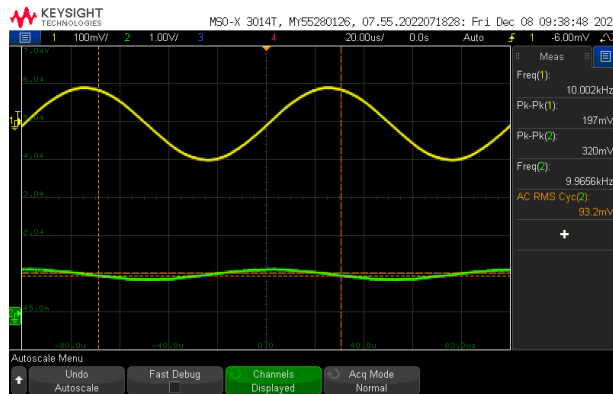


Figure 17: Oscilloscope image for minimum amplification at 10000Hz

This deviation was also evident in our ripple plots. An FRA performed on the amplifier's final voltage output (Figure 18) showed an increase in decibels at higher frequencies. Comparing the RMS voltage at maximum gain frequency (Figure 19) with that at minimum gain frequency (Figure 20) highlighted this discrepancy. According to Equation (7), our amplifier did not maintain a consistent linear profile, with the RMS voltage significantly surpassing the anticipated 15mV.

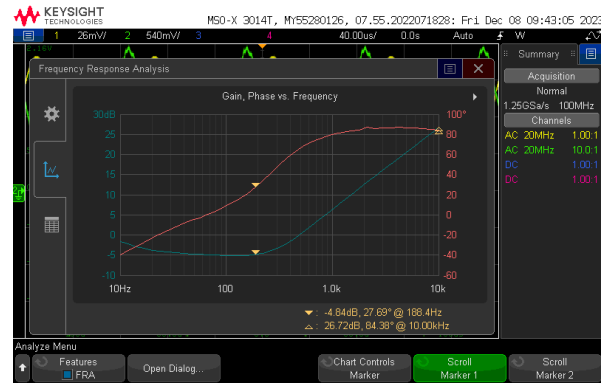


Figure 18: FRA for output voltage of the amplifier

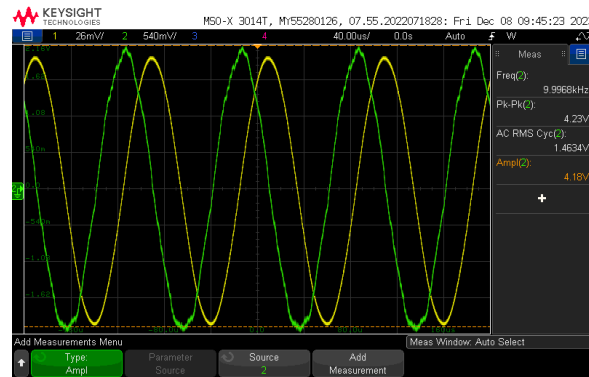


Figure 19: Oscilloscope voltage image for maximum gain frequency

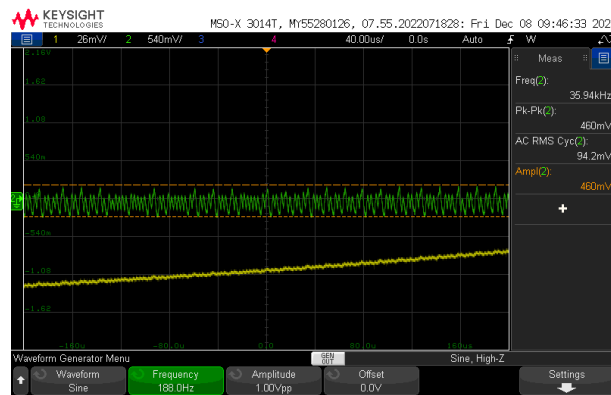


Figure 20: Oscilloscope voltage image for minimum gain frequency

$$\text{Equation (7): } 4.18V - 0.460V = 3.72V \dots > 0.015V$$

Lastly, we evaluated the output power of the amplifier. Equation (8) indicated that the power output was notably higher than the expected 0.4 W.

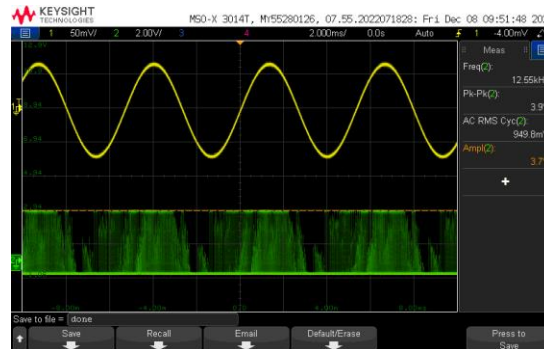


Figure 21: Oscilloscope speaker voltage image for minimum gain frequency

$$\text{Equation (8): } \frac{V_{min}^2}{R_{speaker}} \rightarrow \frac{3.7^2}{8} = 1.71W \dots > 0.4W$$

6. Conclusion

In summary, our project to build an audio equalizer highlighted both successes and areas for improvement. The filters—low pass, high pass, and mid pass—operated effectively within their designated frequency ranges, demonstrating our ability to apply theoretical principles in a practical setting. However, challenges arose with the amplifier, particularly at maximum settings, where the output voltage and power deviated from expected values. These findings underscore the importance of precise calibration and the complexities involved in designing audio circuits. Overall, this project not only reinforced our understanding of electrical engineering concepts but also provided valuable insights into the intricacies of audio signal manipulation and amplification.

7. Sources

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